

Marek ZAREMBA-PIKE, MBCS

marekzp@gmail.com | marekzp.com | UK citizen

Engineering leader specialising in production Generative AI platforms and AI-augmented software delivery. I build self-service systems that let teams define, evaluate, and deploy AI products for millions of users, with the cost, safety, and operational controls needed for five-nines reliability. I also build Photoroom's backend agentic software-development systems, combining coding agents with structured evaluation and deterministic controls, that now produce around 80% of our merged backend PRs.

Technologies

- Python, Django, FastAPI
- Kubernetes, Lambda
- Managed projects in Angular, PHP, React, Vue.js
- AWS, Google Cloud Platform, DigitalOcean & private data centres/on-prem
- Microservices, event-driven arch., data pipelines
- Production applications with Claude, GPT, Gemini, Black Forest Labs, and internal AI APIs
- GitHub Actions, GitLab's CI/CD
- SNS/SQS, custom message queues, AWS Step Functions
- DynamoDB, Elasticsearch, PostgreSQL, Redis
- Datadog, Sentry, Grafana

Recent Work Experience

- 07/2026 – present **Staff Engineer, Engineering Lead Core AI Platform** – Photoroom (AI & ECommerce)
- Designing and building the no-code, self-service platform that enables Photoroom teams to define, test, and launch generative-AI tools.
- Key projects: MCP server, enabling AI assistants to use its image-editing tools; resilient asynchronous execution for long-running AI generations; granular model-cost attribution, analytics, and controls; and AI-safety and content-provenance infrastructure, including child-safety safeguards, C2PA Content Credentials, and imperceptible watermarking.
- Manage three direct reports and lead Photoroom's Backend Guild, setting engineering standards and best practices and promoting knowledge-sharing across all backend engineers.
- 12/2024 – 07/2026 **Staff Engineer, Head of Backend** – Photoroom (AI & ECommerce)
- A hands-on (65% IC) role responsible for Photoroom's Django and FastAPI backend (2,500 requests per second, 10TB database) and managed 4 direct reports
- Oversaw FastAPI AI gateway, handling frontend requests to internal and external AI endpoints, including model fallbacks, rate limiting, abuse prevention, enriching requests, and more to ensure safe and reliable model use at scale
- Expanded our system monitoring (anomalies, SLOs, error rates) to improve incident detection and response times.
- Led migration to Kubernetes, zero-downtime deployments (including data migrations), introduced automated deployments and a testing environment for client developers to use
- Tech lead for cross-platform (Android, backend, iOS, and web) redesign of our billing system to closely align users' AI usage with the subscription fees
- Built an agentic coding pipeline with structured evaluation, delivering 70-90% of merged PRs per week. An increasing majority merged without human review thanks to reference files, and deterministic checks. Pipeline monitored (incl. chain-of-thought) for optimisation.

08/2022 – 10/2024	Staff Engineer – Lantum (Healthcare & Finance) Planned, led, and worked on the refactor of the entire backend (Python/Django), reducing complexity, upgrading libraries, hardening security, and enabling new features. Oversaw tech support and incident response; trained the team, established reporting systems, and built better monitoring and logs. Resulted in reducing exposure of feature squads from 25% of their time to less than 1%, and reduced developer churn Established and evangelised software processes, patterns, and ways of working
07/2021 – 07/2022	Technical Director (CTO) – TrackTrack (LegalTech) Transformed the software architecture, infrastructure, team, and software development processes to allow rapid growth in clients. Some clients required on-prem deployments of our product inside their DMZ.
11/2019 – 06/2021	Consulting Software Engineer & Architect , European Public Affairs Technologies Tech Lead in five successfully delivered projects. I guided clients on the best architecture for their needs, budgets, and maturity levels. I prepared documentation, wrote software, and performed code reviews. Notable projects included:
06/2016 – 12/2019	Backend Python/Django Engineer – European Public Affairs Technologies Built a content sharing platform for EU policymakers and a tool for lobbyists that compiled all the publicly available data on EU politicians.

Example Personal Projects

Savin Hood Tax Calculator

Built a UK income tax calculator to help navigate the 100k tax trap.

<https://app.savinhood.com/calculator>

Zero Downtime Migrations

PostgreSQL migration safety linter for Alembic, Django, and Tortoise.

<https://github.com/Photoroom/zero-downtime-migrations>

Recent Blogs

- [What are PR reviews good for anyway?](#)
- [Senior Claude reviewer is not a good use of engineering talent](#)
- [The laptop is the wrong place to run coding agents](#)

Education

04/2026	AI Safety Bootcamp , ML4Good
09/2024 – ...	MSc Computer Science , University of Bath
06/2024 – 09/2024	AI Safety Fundamentals Alignment Course , BlueDot Impact
03/2024 – 06/2024	AI Programming Nanodegree , Udacity
05/2021 – 10/2021	Engineering Leadership , Cornell University
2010 – 2011	MSc European Public Policy , University College London
2006 – 2010	B.A. Russian Studies & International Relations , University of Birmingham